



Alicja Nowacka

PhD Candidate

IRTG1911 PhD Program

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EDUCATION

11.2020 – 09.2025

University of Lübeck, Germany
Biomedical Research (BMF), Institute for Systemic Inflammation Research ([ISEF](#))
Doctoral studies in **Immunology** (IRTG1911 Program)

09.2022 – 02.2023

Cincinnati Children's Hospital Medical Center, Cincinnati, OH, USA
Division of Immunobiology
Part of doctoral studies (IRTG1911 Program)

2017 – 2020
2014 – 2017

University of Gdansk, Poland
Faculty of **Chemistry**
Department of **Molecular Biotechnology**
Graduate studies
Undergraduate studies

2015 – 2018

Intercollegiate Faculty of Biotechnology University of Gdansk and Medical University of Gdansk
Faculty of **Biotechnology**
Completed by 2nd year

2011 – 2014

Highschool in Bytow, Poland
Biology & Chemistry with advanced English

WORK EXPERIENCE

▪ INTERNSHIPS

09.2019 – 07.2020
09.2017 – 10.2018

▪ **Lund University Research Group**, Sweden
Department of Translational Medicine, Protein Chemistry, Wallenberg Laboratory

09.2018 – 05.2019
09.2016 – 07.2017

▪ **University of Gdansk**, Faculty of Chemistry, Poland
Department of Molecular Biotechnology

07.2017 – 08.2017

▪ **University of Gdansk**, Intercollegiate Faculty of Biotechnology UG&MUG, Poland
Department of Cell Biology and Immunology,

03.2017 – 06.2017

contact person: Assoc. Prof. Marcin Okroj (marcin.okroj@gumed.edu.pl)

01.2017 – 02.2017

Department of Molecular and Cellular Biology, Laboratory of Protein Biochemistry

Department of Biotechnology, Laboratory of Plant Protection and Biotechnology

07.2017
08.2016 – 09.2016

▪ **Department of Veterinary Hygiene**, Gdansk, Poland

▪ **BioVentures Institute**, Gdansk, Poland

▪ SCIENTIFIC CONFERENCES

04.2025

▪ Arbeitskreis Komplementsystem ([AKKS](#)) Meeting 2025, Hamburg, Germany
Oral presentation

01.2025

▪ Obergurgl Retreat of the EU ITN [CORVOS](#) and [HOROS](#), Obergurgl, Austria
Oral presentation

09.2024

- 19th European Meeting on Complement in Human Disease ([EMCHD](#)), Lübeck, Germany
Poster presentation (P97) + **Award**; [Abstract](#)

04.2024

- Arbeitskreis Komplementsystem ([AKKS](#)) Meeting 2024, Tutzing, Germany
Oral presentation

01.2024

- Obergurgl Retreat of the EU ITN [CORVOS](#) and [HOROS](#), Obergurgl, Austria
Oral presentation

09.2023

- 29th International Complement Workshop ([ICW](#)), Newcastle, England
Oral presentation + **Travel award**; [Abstract](#)

10.2022

- Immunology retreat, Cincinnati, Ohio, USA
Poster presentation

08.2022

- 18th European Meeting on Complement in Human Disease ([EMCHD](#)), Bern, Switzerland
Oral presentation + **Travel award**; [Abstract](#)

03.2022

- Arbeitskreis Komplementsystem ([AKKS](#)) Meeting 2022, Tutzing, Germany
Oral presentation

03.2021

- Arbeitskreis Komplementsystem (AKKS) Meeting 2021, online (due to the pandemic)
Oral presentation

05.2019

- 8th Intercollegiate Biotechnology Symposium "[Symbioza](#)" Warsaw, Poland
Oral presentation

12.2016

- Co-organizer of medical conferences, [Via Medica](#), Sopot, Poland

10.2016

- Heart and Vessels Diseases
- Connected with arterial hypertension

▪ LIST OF PUBLICATIONS

*Manuscript
in preparation*

Nowacka A. A., Einwohlt P., Ohms M., Geisler S., Abdala Y., Fähnrich A., Köhl J. *C3a and C5a regulate DC progenitor homeostasis and their differentiation into cDC1 and cDC2 subsets*

11.2025

Nowacka A. A., Sordo Vieira L., Petr V., Fageräng B., Würzner R., Ohms M. *The grand escape – how pathogens outsmart the human complement system*. Immunobiol. 2025 230:153126 <https://doi.org/10.1016/j.imbio.2025.153126>

06.2023

Laumonnier Y., Korkmaz R. U., **Nowacka A. A.**, Köhl J. *Complement-mediated immune mechanisms in allergy*. Eur. J. Immunol. 2023 53: 2249979 <https://doi.org/10.1002/eji.202249979>

05.2022

Kremlitzka, M., Colineau L., **Nowacka A. A.**, Mohlin F. C., Wozniak K., Blom A. M., King B. C. *Alternative translation and retrotranslocation of cytosolic C3 that detects cytoinvasive bacteria*. Cell. Mol. Life Sci. 2022 79: 291 <https://doi.org/10.1007/s00018-022-04308-z>

03.2019

Kremlitzka, M., **Nowacka A. A.**, Mohlin F. C., Bompada P., Marinis Y. D., Blom A. M. *Interaction of Serum-Derived and Internalized C3 with DNA in Human B Cells – A Potential Involvement in Regulation of Gene Transcription*. Front. Immunol. 2019 10:493 <https://doi.org/10.3389/fimmu.2019.00493>

▪ PARTICIPATING IN PROJECTS

**University of Lübeck, Institute for Systemic Inflammation Research
Research Group of Prof. Dr. med. Jörg Köhl**

PhD project:

Anaphylatoxin-mediated regulation of conventional dendritic cell differentiation and function

supervisor: Prof. Dr. med. Jörg Köhl (joerg.koehl@uksh.de)

**Cincinnati Children's Hospital Medical Center, Cincinnati, OH, USA
Division of Immunobiology
Research Group of Prof. DVM. Chandrashekhar Pasare**

PhD project:

Role of C5aR1 in antigen and TLR-driven T cell proliferation and differentiation by splenic conventional dendritic cells

supervisor: Prof. DVM. Chandrashekhar's Pasare (Chandrashekhar.Pasare@cchmc.org)

**Lund University, Department of Translational Medicine, Protein Chemistry
Research Group of Prof. Anna Blom**

11.2020 – 09.2025

09.2022 – 02.2023

Projects during internships:

09.2019 – 07.2020

Investigating intracellular roles of C3 in autophagy and pathogen detection

supervisors: Prof. Anna Blom, Assoc. Prof. Ben King, PhD Lucie Colineau

contact person: Prof. Anna Blom (anna.blom@med.lu.se), Assoc. Prof. Ben King (ben.king@med.lu.se)

09.2017 – 10.2018

Investigating the role of C3 and C3aR in human B cells

supervisors: Prof. Anna Blom, PhD Mariann Kremlitzka

contact person: Prof. Anna Blom (anna.blom@med.lu.se)

University of Gdansk, Department of Molecular Biotechnology
Prof. Piotr's Skowron research group

Master's project:

09.2018 – 07.2019

Biosynthesis and purification of recombinant TRFH domains responsible for the homodimerization of human TRF proteins and proper telomeres function

supervisor: Assoc. Prof. Agnieszka Zylicz-Stachula

Bachelor's project:

08.2016 – 07.2017

Isolation and purification of recombinant polyepitopic protein S_5 – the potential vaccine against hepatitis B

supervisors: Assoc. Prof. Agnieszka Zylicz-Stachula, PhD Joanna Zebrowska

contact person (both): agnieszka.zylicz-stachula@ug.edu.pl

▪ AWARDS

09.2024

ECN Poster Prize 2024 awarded by the European Complement Network Committee on the occasion of the 19th European Meeting on Complement in Human Disease for an excellent poster (Lubeck, Germany)

09.2023

ICW 2023 Travel Award awarded by Local Organising Committee and the International Complement Society ([ICS](https://www.icw2023.org/)) and the ICS Council for outstanding scientific contribution to ICW2023 (Newcastle, England)

08.2022

EMCHD 2022 Travel Award awarded by Committee of the 18th European Meeting in Complement for Human Disease (Bern, Switzerland) for an oral presentation entitled *Anaphylatoxin receptors regulate the differentiation and function of dendritic cells subsets from bone marrow-derived precursors*

09.2019 – 07.2020

09.2017 – 10.2018

Scholarship from Lund University for students

2016 – 2017

Best Students of the University of Gdansk, Faculty of Chemistry (XIII Edition)

2017 – 2018

Scholarship for Academic Merit from University of Gdansk

2016 – 2017

2015 – 2016

▪ LICENCES AND CERTIFICATES

09.2022

CITI Program

- Working with the IACUC
- Working with Mice in Research Settings
- Lab Core Curriculum

10.2020

Laboratory Animal Science Course – Category **FELASA B**

▪ METHODOLOGICAL COMPETENCE

In vivo mouse models

Cell biology

Intraperitoneal injection

Isolation and purification of various murine immune cells from lung tissue, bone marrow, spleen, blood, tracheas, brain

In vitro differentiation of bone marrow-derived dendritic cells

In vitro co-culture system of MDP-derived cDC (CD172a⁻ or CD172a⁺) and T cells

Preparation of electrocompetent cells

Functional assays

Molecular biology	RNA extraction cDNA preparation Real-Time PCR DNA isolation and purification Restriction digestion and analysis of DNA Molecular cloning (gene cloning)
Protein biology	Bacterial expression of recombinant proteins Protein isolation and purification Cell lysis SDS polyacrylamide gel electrophoresis (SDS-PAGE) Western blot Flow cytometry ELISA
Immunohistochemistry	Chromatography techniques: affinity, ion exchange, size exclusion (gel filtration), TLC Fluorescence microscopy

■ ADDITIONAL INVOLVEMENT

06.2019 – 06.2020

President at BEST Lund (Board of European Students of Technology; non-profit, non-political organization) Sweden; acquired skills: leadership, facilitation the meetings, delegating tasks, motivational skills, time management, communication skills

01.2017 – 06.2020

Member of BEST Lund, Sweden; PR responsibilities: promotion of the events, design posters, T-shirts, graphics, promotion materials; co-organizer of Summer Course 2018 and 2019 in Lund, co-organizer of Local Round EBEC (European BEST Engineering Competition) 2018 in Lund

10.2018 – 10.2019

Guest member of BEST Gdansk, Poland; PR responsibilities: promotion of the events, graphic design of the leaflets; co-organizer of Summer Course 2019 in Gdansk: HR Participants Responsible

03.2019

Participant of beKNOWN – Marketing and public relations training, organized by **BEST Iasi**, Romania

2017 – 2019

Member of Boost Biotech, Gdansk, Poland (nonprofit bio – tech organization), PR responsibilities: photo reports of the events

2016 – 2017

Students' Conference and Research Grants Funding Committee Member (Chemistry Department, University of Gdansk, Poland)

2016 – 2017

Faculty Board Member (Faculty of Chemistry, University of Gdansk, Poland)

■ SKILLS

Programs:

- MS Office
- GraphPad Prism
- FlowJo
- Adobe Illustrator, Photoshop, Premiere Rush
- Affinity: Design, Photo

■ PERSONAL FEATURES

- Leadership
- Self-motivated
- Problem solver
- Motivated
- Hard-working
- Communicative
- Responsible
- Creative
- Team worker
- Fast learner

■ LANGUAGES

Polish	Native speaker
English	Advanced understanding of both written and spoken
German	Intermediate understanding of both written and spoken
Swedish	Intermediate understanding of both written and spoken

■ INTERESTS

Travelling, learning about cultures and customs of other countries, psychology, personal development, languages, photography, graphics design, rock climbing, playing the piano, baking, cooking